# EE4047 – High Voltage Techniques I

## Continuation to:

## Breakdown in Solids

**Week 14.2 – Thursday Lesson**





**Asst. Prof. Dr. M. Alparslan**

**08.06.2023**

# OUTLINE

1 Breakdown of solid insulating materials

2 Losses in solid dielectrics

3 Breakdown types in solid dielectrics

# 1 BREAKDOWN OF SOLID INSULATION MATERIALS

**Solid dielectrics**

- Ceramic
- Glass
- Silicones
- Paper, Oiled paper
- Resin

# 1 BREAKDOWN OF SOLID INSULATION MATERIALS

In solid dielectrics, highly purified and free of imperfections, the breakdown strength is high, of the order of 10 MV/cm.

The highest breakdown strength obtained under carefully controlled conditions is known as the ‘’intrinsic strength’’ of the dielectric.

# LOSSES IN SOLID DIELECTRICS

## Losses in solid dielectrics

- Dielectric losses
- Dipole losses
- Hysteresis losses
- Ionization losses

The common outcome, **material heats up**

# 3 LOSSES IN SOLID DIELECTRICS

## Breakdown types in solid dielectrics

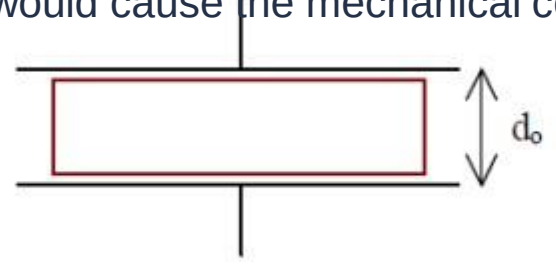
- Instantaneous breakdown
- Electromechanical breakdown
- Electrical breakdown
- Cavity discharge (internal discharge)
- Treeing
- Thermal breakdown

# LOSSES IN SOLID DIELECTRICS

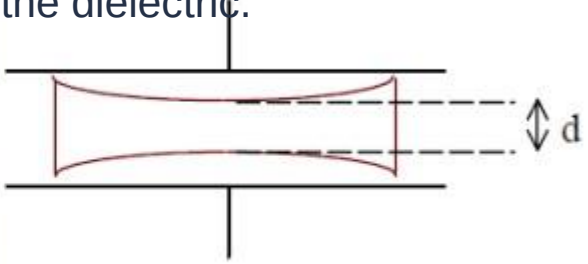
## Electromechanical breakdown

When an electric field is applied to a dielectric between two electrodes, a mechanical force will be exertes on the dielectric due to the force of attraction between the surface charges. This can be also due to deformation of the structure of the insulating material by time during operation.
This compression decreases the dielectric thickness, thus increasing effective stress.

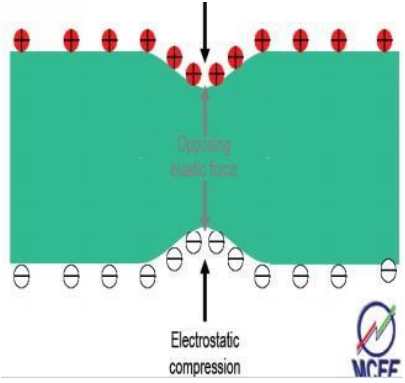
When the field is increased, the thickness of the material decreases. Increase in the field may would cause the mechanical collapse of the dielectric.




In the absence of field




With applied field

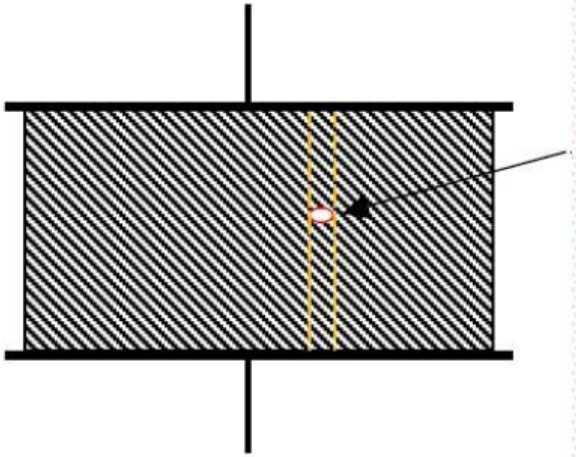


## Breakdown due to internal discharges

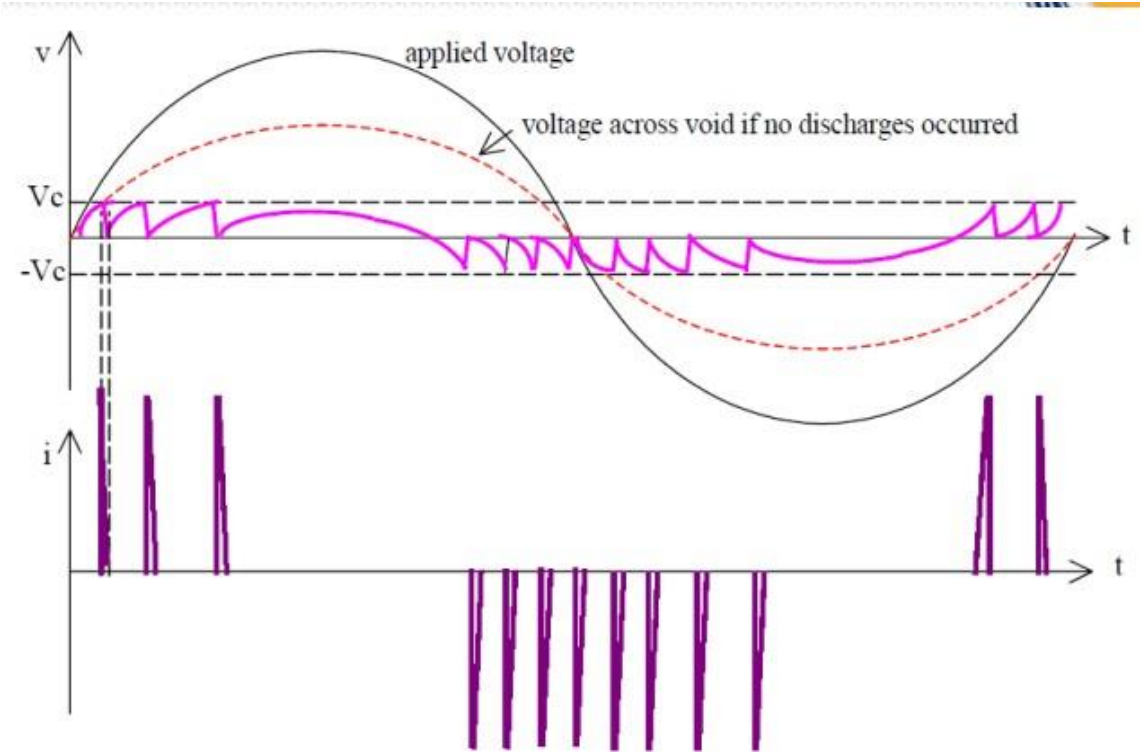
Solid insulating materials sometimes contain voids or cavities in the medium or boundaries between the dielectric and the electrodes.
These voids have a dielectric constant of unity and a lower dielectric strength.

Hence the electric field strength in the voids is higher than that across the dielectric.

Thus even under normal working voltages, the field in the voids may exceed their breakdown value and breakdown may occur.



## Breakdown due to internal discharges



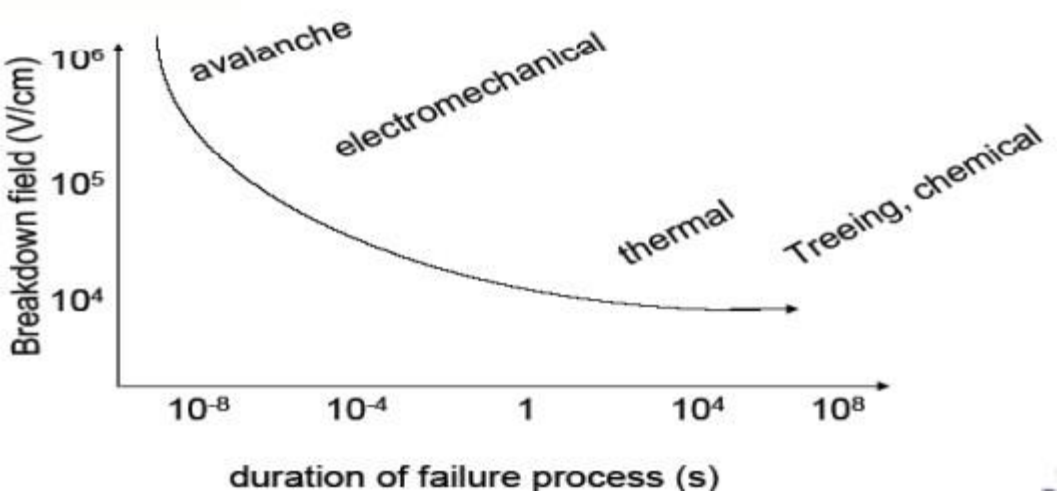
## Thermal breakdown

Heat is generated continuously in electricaly stressed insulation by dielectric losses, which is transferred to the surrounding medium by conduction through the solid dielectric and by radiation from its outer surfaces.

If the heat generated exceeds the heat lost to the surroundings, the temperature of the insulation increases.

# LOSSES IN SOLID DIELECTRICS

Time and field of failure of different breakdown mechanisms



# END OF THE LESSON

## Any questions?

Check out the following videos on Youtube:

Breakdown voltage testing of transformer oil: https://www.youtube.com/watch?v=nZAg_A1tb0o

Animation of transformer oil testing: https://www.youtube.com/watch?v=BnrARAP2btk

Transformer oil test on site training: https://www.youtube.com/watch?v=YTp-SjvaA5Q

Oil testing equipment video: https://www.youtube.com/watch?v=WxVKADaLyXk